



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

16-06-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	A	A	B	A	C	D	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	A	B	B	C	A	D	D	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	25	37	3	187	1500	9	4	1	5

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	B	B	B	B	C	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	A	C	B	A	D	D	B	D	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	9	4	3	7	3	8	3	4	2	9

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	C	B	D	D	C	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	B	A	C	B	A	A	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7	1	5	3	49	6	3	150	2	8

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (A)**

Work done on 4 C charge

$$= \int q\vec{E} \cdot d\vec{r} = q \int_{\sqrt{2}}^3 E dr$$

$$[\because r \text{ for } (1, 1, 0) = \sqrt{2}]$$

$$\& r \text{ for } (3, 0, 0) = 3]$$

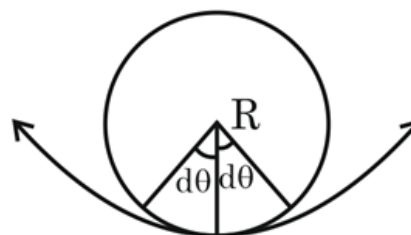
$$= 4 \times \text{area of E-r graph from}$$

$$r = \sqrt{2} \text{ m to } r = 3$$

$$= 4 \times \left[\frac{1}{2} (3 - \sqrt{2}) 20 \right] = 40(3 - \sqrt{2}) \text{ J}$$

2. **Ans (B)**

$$B \text{ at end} = \frac{1}{2} B \text{ at interior} = \frac{1}{2} B$$



$$2IdL \left(\frac{B}{2} \right) = 2T \sin \theta$$

$$dL = R(2d\theta)$$

$$2I R \cdot 2d\theta \frac{B}{2} = 2T d\theta$$

$$T = BIR$$

3. Ans (A)

Electric field on surface of a uniformly charged sphere is given by

$$\frac{Q}{4\pi\epsilon_0 r^2} = \frac{\rho R}{3\epsilon_0}$$

Electric field at outside point is given by

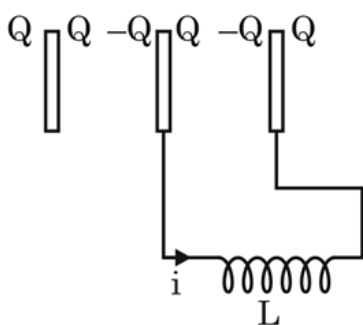
$$E = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{\rho R^3}{3\epsilon_0 r^2}$$

$$|\vec{E}_r| = \frac{\rho r_0}{3\epsilon_0} - \frac{\rho \left(\frac{r_0}{2}\right)^3}{3\epsilon_0 \left(\frac{3r_0}{2}\right)^2}$$

$$= \frac{17\rho r_0}{54\epsilon_0} \text{ left}$$

4. Ans (A)

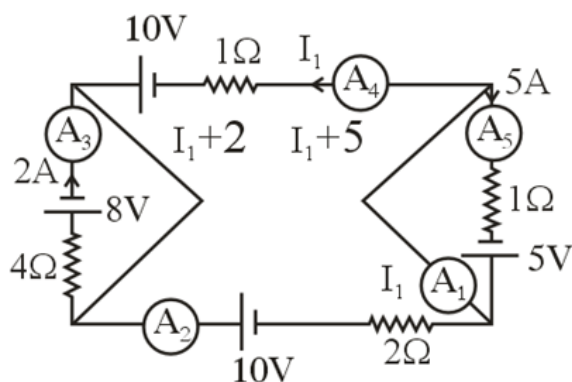
$$\frac{Q^2}{2C} = \frac{1}{2} Li_0^2$$



$$\Rightarrow i_0 = \frac{Q}{\sqrt{LC}}$$

5. Ans (B)

$$-I_1 + 10 - 10 - 2I_1 = 0$$



$$I_1 = 0$$

6. Ans (A)

Induced e.m.f. = - rate of change of magnetic flux

Inside the solenoid, flux = B × Area

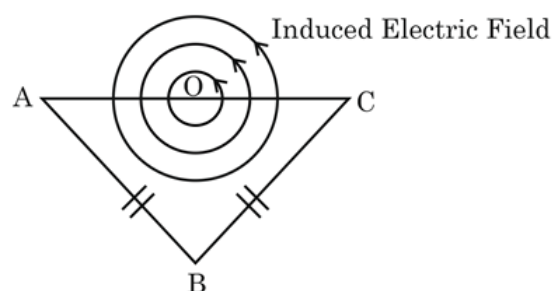
where B is the magnetic flux density

induced e.m.f.

$$= -\frac{d}{dt}(B \times \text{Area}) = -\text{Area} \times \frac{dB}{dt} = -R^2 \frac{dB}{dt}$$

since area of triangle ABC is R^2 .

(b) then important point is that a changing magnetic field induces a circulating electric field (from which we derive the induced e.m.f.). in other words, there are no radial components of electric field (radiating outwards or inwards) so that no e.m.f. is actually induced in the length AC of the circuit.



Hence the induced e.m.f. is shared out between AB and BC.

$$\text{So for AB, induced e.m.f} = -\frac{1}{2} R^2 \frac{dB}{dt}$$

7. Ans (C)

There are four semicircular rod each of length πr

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$

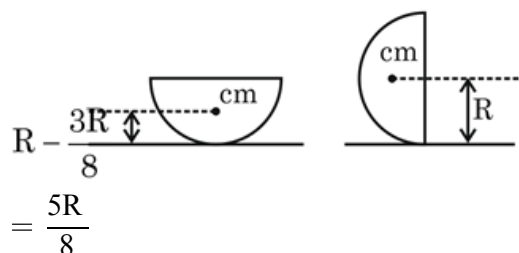
$$\frac{1}{R_{eq}} = \frac{KA}{\pi r} + \frac{KA}{\pi r} + \frac{KA}{\pi r} + \frac{KA}{\pi r}$$

$$\frac{1}{R_{eq}} = \frac{4KA}{\pi r}$$

$$R_{eq} = \frac{\pi r}{4KA}$$

8. **Ans (D)**

When it is released it slides on ground



from conservation of mechanical energy

$$U_i + K_i = U_f + K_f$$

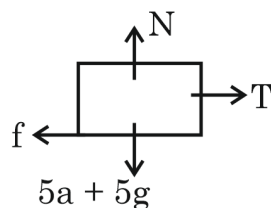
$$mgR + 0 = mg \left(\frac{5R}{8} \right) + \frac{1}{2}mv^2$$

$$\frac{3mgR}{8} = \frac{1}{2}mv^2$$

$$v = \frac{\sqrt{3gR}}{2}$$

9. **Ans (A)**

FBD of 5kg block



$$T \geq f \quad \dots(i)$$

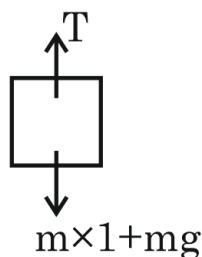
$$N = 5 \times 1 + 5g \quad \dots(ii)$$

$$N = 55N$$

$$f = \mu (5 \times 1 + 5g)$$

$$f = \frac{2}{10}(55) = 11N$$

FBD of m block



$$\Rightarrow T = m + mg \quad \dots(iii)$$

from equation (i) & (iii)

$$m + mg \geq f$$

$$m(1 + 10) \geq 11$$

$$m \geq 1kg$$

10. **Ans (A)**

$$\vec{r}_{\text{com}}(\text{rod}) = \frac{L}{2} \hat{j}$$

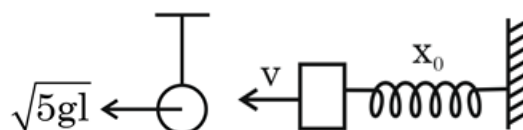
$$\vec{r}_{\text{com}}(\text{semicircular ring}) = \frac{2 \left(\frac{L}{4} \right)}{\pi} \hat{i} + \frac{3L}{4} \hat{j}$$

$$\vec{r}(\text{system}) = \frac{M\vec{r}_{\text{com}}(\text{rod}) + M\vec{r}_{\text{com}}(\text{ring})}{M + M}$$

$$\vec{r}(\text{system}) = \frac{1}{2} \left(\frac{L}{2} \hat{j} + \frac{L}{2\pi} \hat{i} + \frac{3L}{4} \hat{j} \right)$$

$$\vec{r}(\text{system}) = \frac{L}{4\pi} \hat{i} + \frac{5L}{8} \hat{j}$$

11. **Ans (A)**



$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2; v = \sqrt{\frac{k}{m}} \cdot x$$

since the collision is perfectly elastic & mass of bob & block are same, velocity gets exchanged.

Hence bob will start with $v = \sqrt{\frac{k}{m}}x$ & block will come at halt.

$$v_{\text{bob}} = \sqrt{\frac{k}{m}}x = \sqrt{5gl} \quad (\text{condition for looping the loop})$$

$$x = \sqrt{\frac{5glm}{k}} = \sqrt{\frac{5 \times 10 \times 1 \times 0.1}{100}}$$

$$x = 0.22m$$

12. **Ans (C)**

Applying law conservation of angular momentum

about centre of the disc

$$\frac{1}{2}MR^2\omega = \left(\frac{1}{2}MR^2 + \mu tR^2 \right) \frac{\omega}{4}$$

$$\frac{M}{2} = \frac{M}{8} + \frac{\mu t}{4} \quad (\mu t = \text{mass of water absorbed by cotton pad after time 't'})$$

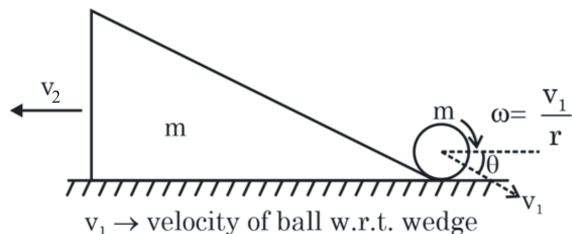
$$\frac{3M}{8} = \frac{\mu t}{4}$$

$$t = \frac{3M}{2\mu}$$

13. Ans (A)

As no ext. force in horizontal direction is acting on system, here we use by conservation of momentum

$$mv_2 = m(v_1 \cos \theta - v_2) \Rightarrow v_1 = \frac{2v_2}{\cos \theta}$$



By work energy theorem-

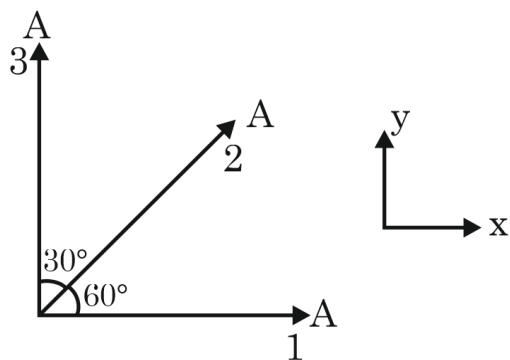
$$mgh = \frac{1}{2}mv_2^2 + \frac{1}{2}m \left[(v_1 \cos \theta - v_2)^2 + (v_1 \sin \theta)^2 \right] + \frac{1}{2} \left(\frac{2}{5}mr^2 \right) \left(\frac{v_1}{r} \right)^2$$

$$2gh = v_2^2 + v_1^2 + v_2^2 - 2v_1 v_2 \cos \theta + \frac{2}{5}v_1^2$$

$$2gh = 2v_2^2 + \frac{7}{5} \left(\frac{4v_2^2}{\cos^2 \theta} \right) - 4v_2^2$$

$$\Rightarrow v_2 = \sqrt{\frac{5gh}{14\sec^2 \theta - 1}} = \frac{10}{9}$$

14. Ans (B)



If R is the resulting amplitude of motion

$$R_x = A + A \cos 60^\circ = A + \frac{A}{2} = \frac{3A}{2}$$

$$R_y = A + A \cos 30^\circ = A + \frac{\sqrt{3}A}{2} = A \left(1 + \frac{\sqrt{3}}{2} \right)$$

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{\frac{9A^2}{4} + A^2 \left(1 + \frac{\sqrt{3}}{2} \right)^2}$$

$$R = A\sqrt{4 + \sqrt{3}}$$

15. Ans (B)

$$a_t = 0, a_n = \frac{v^2}{r}$$

Radius of curvature on particle trajectory is

$$r = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\left| \frac{d^2y}{dx^2} \right|}$$

At x = 1

$$\frac{dy}{dx} = \sqrt{3}, \frac{d^2y}{dx^2} = \sqrt{3}$$

$$\text{so } r = \frac{8}{\sqrt{3}}$$

$$\Rightarrow a_N = \frac{v^2}{r} = \frac{\sqrt{3}}{2} m/s^2$$

16. Ans (C)

The mass of element is $dm = \frac{m}{L} \cdot R d\theta$

tangential force on element is $dF = dm g \sin \theta$

Total tangential force on chain

$$F_t = \int dF = \int_0^{L/R} \frac{m}{L} R g \sin \theta d\theta$$

$$F_t = \frac{mgR}{L} [-\cos \theta]_0^{L/R}$$

$$F_t = \frac{mgR}{L} \left(1 - \cos \left(\frac{L}{R} \right) \right)$$

tangential acceleration of chain

$$a_t = \frac{F_t}{m} = \frac{gR}{L} \left[1 - \cos \left(\frac{L}{R} \right) \right]$$

17. Ans (A)

Sound velocity in a gaseous medium is given as

$$v = \sqrt{\frac{B}{\rho}} \text{ where } B \rightarrow \text{Bulk modulus of medium}$$

$$B = -v \frac{dP}{dv}$$

$$\text{in the given } P = \frac{\alpha}{v^2}$$

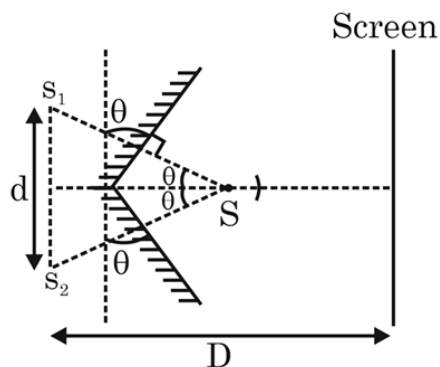
$$dP = -\frac{2\alpha}{v^3} dv$$

$$B = -v \left(-\frac{2\alpha}{v^3} \right) = \frac{2\alpha}{v^2} = 2P$$

$\Rightarrow$ in air sound velocity is given as

$$v = \sqrt{\frac{2P}{\rho}}$$

18. Ans (D)



In figure shown distances $SS_1 = SS_2 = 2a \cos \theta$
separation between S_1 and S_2 is
 $d = 2(2a \cos \theta) \sin \theta = 2a \sin 2\theta = 4a\theta$ [as θ is small]
separation between slit plane ($S_1 S_2$) and screen
 $D = (2a \cos \theta) \cos \theta + b$ [for small $\theta \cos \theta = 1$]
 $D = 2a + b$
fringe width in interference pattern
 $B = \frac{\lambda D}{d} = \frac{\lambda(2a + b)}{4a\theta}$

19. Ans (D)

Initially p.d. across capacitors

$$V_1 = \frac{2V}{3}; V_2 = \frac{V}{3}$$

$$U_{i(2c)} = \frac{1}{2}(2c)\left(\frac{V}{3}\right)^2 = \frac{cV^2}{9}$$

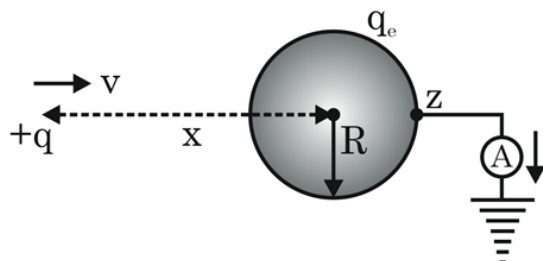
after insertion of dielectric, p.d. across capacitors

$$V_1 = \frac{10V}{11}; V_2 = \frac{V}{11}$$

$$U_{f(2c)} = \frac{1}{2}(10c)\left(\frac{V}{11}\right)^2 = \frac{5cV^2}{121}$$

$$\frac{U_{i(2c)}}{U_{f(2c)}} = \frac{\frac{cV^2}{9}}{\frac{5cV^2}{121}} = \frac{121}{45}$$

20. Ans (A)



Here $V_z = 0$

$$\frac{Kq}{x} + \frac{Kq_e}{R} = 0$$

$$\Rightarrow q_e = -\frac{qR}{x}$$

As x is continuously decreasing, charge on ball must be continuously increasing hence ammeter shows a current i .

$$i = \frac{dq_e}{dt} = \frac{qR}{x^2} \frac{dx}{dt}$$

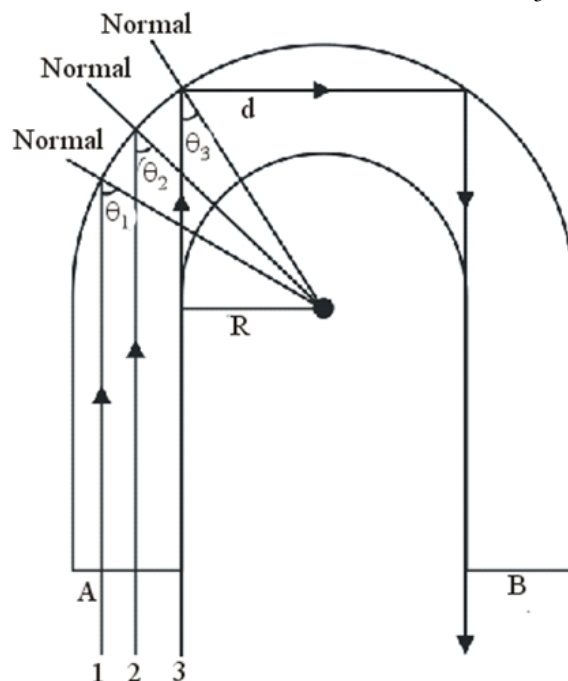
$$\Rightarrow i = \frac{qR}{x^2} v$$

PART-1 : PHYSICS

SECTION-II

1. Ans (5)

Figure shows three rays 1, 2, 3 incident on plane face A. We can see that angle of incidence at curved surface is least for ray 3. If ray 3 reflects at the curved surface, then all the rays will reflect as their angle of incidence is greater than angle θ_3 .



Hence the required condition is

$$\theta_3 \geq \text{critical}$$

$$\text{As } \sin \theta_{\text{critical}} = \frac{1}{\mu} = \frac{2}{3}$$

from geometry of figure, we have

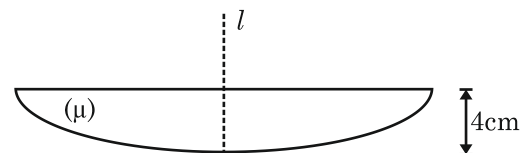
$$\sin \theta_3 = \frac{R}{R + d}$$

$$\text{So } \sin \theta_3 \geq \sin \theta_{\text{critical}}$$

$$\frac{R}{d + R} \geq \frac{2}{3} \text{ or } \frac{d}{R} \leq \frac{1}{2}$$

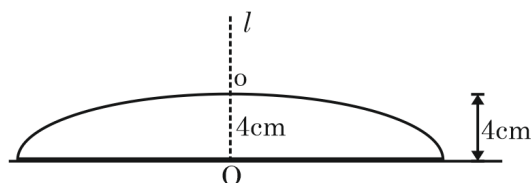
$$\text{Therefore } \left(\frac{d}{R}\right)_{\text{maximum}} = \frac{1}{2}$$

2. Ans (25)



$$\text{App depth } 3\text{cm} = \frac{4\text{cm}}{\mu} \Rightarrow \mu = \frac{4}{3}$$

we use refraction formula for apparent depth as-



$$\mu_1 = \frac{4}{3}; u = -4\text{ cm}; R = -R$$

$$\mu_2 = 1; v = -\frac{25}{8}\text{ cm}$$

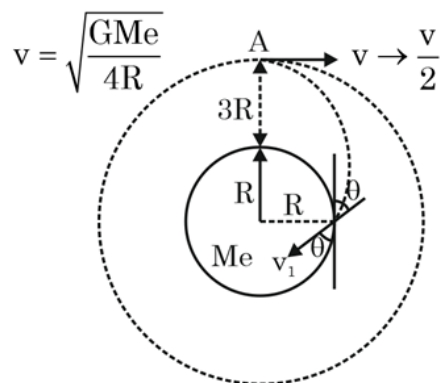
$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$-\frac{8}{25} + \frac{4}{3 \times 4} = \frac{1 - \frac{4}{3}}{-R}$$

$$\frac{1}{3R} = \frac{1}{3} - \frac{8}{25} = \frac{1}{75}$$

$$R = 25\text{ cm}$$

3. Ans (37)



With respect to earth, we conserve energy and angular momentum of satellite as-
by angular momentum-

$$m \left(\frac{v}{2} \right) 4R = m(v_1 \cos \theta) R$$

$$2\sqrt{\frac{GMe}{4R}} = v_1 \cos \theta \dots (1)$$

$$\frac{1}{2} m \left(\frac{v}{2} \right)^2 - \frac{GMe m}{4R} = \frac{1}{2} m v_1^2 - \frac{GMe m}{R} \dots (2)$$

$$v_1 = \frac{5}{4} \sqrt{\frac{GMe}{R}}$$

from (1)

$$\sqrt{\frac{GMe}{R}} = \frac{5}{4} \sqrt{\frac{GMe}{R}} \cos \theta$$

$$\cos \theta = \frac{4}{5} \Rightarrow \theta = 37^\circ$$

4. Ans (3)

For equilibrium of lighter sphere

$$T + v(0.5\rho_w)g = V\rho_w g$$

$$T = 2.565\text{ N}$$

for equation of heavier sphere

$$T + N + V\rho_w g = V(2\rho_w)g$$

$$N = 2.565\text{ N}$$

5. Ans (187)

Rate of heat conduction through bottom of boiler is

$$\frac{dQ}{dt} = \frac{kA(T_2 - T_1)}{x} = \frac{63 \times 1 \times (300 - 100)}{3 \times 10^{-2}}$$

$$= 4.2 \times 10^5 \text{ J/sec}$$

If rate of evaporate of water is m kg/sec, we can use

$$\frac{dQ}{dt} = mL$$

$$m = \frac{dQ/dt}{L} = \frac{4.2 \times 10^5}{2251.2 \times 10^3} = 0.1865 \text{ kg/sec}$$

6. Ans (1500)

Change in internal energy $\Delta U = nC_v (T_B - T_A)$

$$= \frac{3R}{2} (T_B - T_A)$$

for point (state of gas) B, we use $TV^{\gamma-1}$ constant to state C

$$\Rightarrow T_C V_C^{\gamma-1} = T_B V_B^{\gamma-1}$$

$$\Rightarrow T_0 V_0^{\gamma-1} = T_B (2V_0)^{\gamma-1} \Rightarrow T_B = \frac{T_0}{2^{\gamma-1}} = \frac{T_0}{2^{\frac{5}{3}-1}}$$

$$= \frac{T_0}{2^{\frac{2}{3}}}$$

[As $\gamma = \frac{5}{3}$ for monoatomic gas]

$$\Rightarrow \Delta U = \frac{3R}{2} \left(\frac{T_0}{2^{\frac{2}{3}}} - T_0 \right) = \frac{3RT_0}{2} \left(\frac{1 - 2^{\frac{2}{3}}}{2^{\frac{2}{3}}} \right)$$

$$= -1500\text{ J}$$

7. **Ans (9)**

Amplitude of oscillations at a distance x from clamped end is

$$R = A_0 \sin kx$$

$$R = A_0 \sin \left[\frac{2\pi}{(2\ell/3)} \cdot x \right] = A_0 \sin \left(\frac{3\pi x}{\ell} \right)$$

For II overtone, freq. of oscillations string

$$n = \frac{3v}{2\ell} = \frac{3}{2\ell} \sqrt{\frac{T}{\mu}}$$

for an element of string of width dx , max KE is given as

$$dE = \frac{1}{2} (\mu dx) \omega^2 R^2$$

$$= \frac{1}{2} \mu \omega^2 A_0^2 \sin^2 \left(\frac{3\pi x}{\ell} \right) dx$$

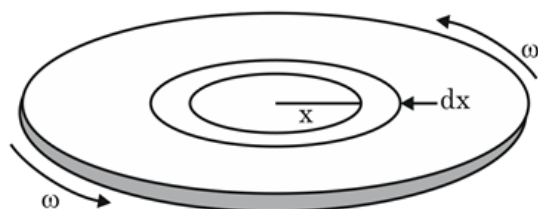
$$E = \int dE = \frac{9\pi^2 T A_0^2}{2\ell^2} \int_0^\ell \sin^2 \left(\frac{3\pi x}{\ell} \right) dx$$

$$E = \frac{9\pi^2 A_0^2 T}{4\ell} = 9J$$

8. **Ans (4)**

If we consider on elemental ring on disc as shown, the speed particles of lubricant just below this element is

$$v = x\omega$$



velocity gradient in lubricant below this element is

$$\frac{dv}{dx} = \frac{x\omega}{t}$$

viscous force on element (tangentially) is

$$dF = \eta (2\pi x dx) \frac{x\omega}{t}$$

torque of viscous force on element is

$$d\tau = dF \cdot x = \frac{2\pi\eta\omega}{t} x^3 dx$$

$$\text{Total torque on disc } \tau = \int d\tau = \frac{2\pi\eta\omega}{t} \int_0^{d/2} x^3 dx$$

$$= \frac{2\pi\eta\omega}{t} \left[\frac{x^4}{4} \right]_0^{d/2} = \frac{\pi\eta\omega d^4}{32t} = 0.4N\cdot m$$

9. **Ans (1)**

To produce 32MW power, number of fission required/sec are

$$N = \frac{32 \times 10^6}{200 \times 10^6 \times 1.6 \times 10^{-19}} = 10^{18} \text{ fission/sec}$$

for 1000 hour of operation, number of total fission required are

$$N_T = 10^{18} \times 1000 \times 3600 = 3.6 \times 10^{24} \text{ fission}$$

$$\text{mass of } ^{235}\text{U required is } m = \frac{3.6 \times 10^{24}}{6.023 \times 10^{23}} \times 235$$

$$= 1404.6 \text{ gm}$$

$$= 1.41 \text{ kg}$$

10. **Ans (5)**

$$\text{Energies of emitted photons } \Delta E_1 = \frac{12431}{1085}$$

$$= 11.4 \text{ eV}$$

$$\&\Delta E_2 = \frac{12431}{304} = 40.9 \text{ eV}$$

Total de. excitation energy is $\Delta E_T = \Delta E_1 + \Delta E_2$

$$= 11.4 + 40.9 = 52.3 \text{ eV}$$

If He^+ is excited to n^{th} energy level

$$\Delta E = 13.6(z)^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$52.3 = 13.6 \times 4 \left[1 - \frac{1}{n^2} \right]$$

$$\Rightarrow n^2 = 25 \Rightarrow n = 5$$

PART-2 : CHEMISTRY

SECTION-I

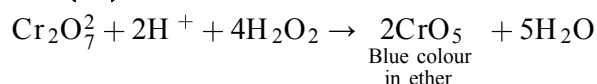
1. **Ans (C)**

Broad spectrum antibiotics act on different antigens.

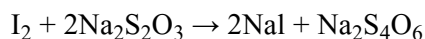
3. **Ans (A)**

Hybrid Propellant – A hybrid propellant consists of a solid fuel and liquid oxidizer to provide propulsion energy and working substance e.g. Solid acrylic rubber and liquid N_2O_4

4. **Ans (B)**



6. **Ans (B)**



Moles of I_2 liberated = 1.7 m mole

Moles of Cl in bleaching powder = 3.4 m mole
= 0.12 gm Cl

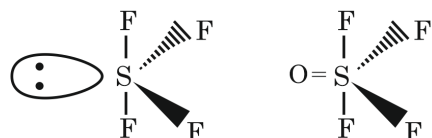
$$\% \text{ of Cl} = \frac{0.12}{0.6} \times 100 = 20$$

7. **Ans (B)**

Bond angle

$\text{H}_2\text{O} = 104.5^\circ$, $\text{NH}_3 = 107^\circ$, $\text{OF}_2 = 103^\circ$, $\text{NF}_3 = 102^\circ$

8. **Ans (C)**



axial FSF angle	179	178.5
equatorial FSF angle	103	110

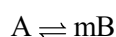
9. **Ans (B)**

(b) A collides with A in an elementary reaction is most likely the scenario here.

(c) Again a ter-molecular collision is almost impossible.

(d) Reactant 'B' is missing from rate law indicating a complex reaction.

10. **Ans (A)**



$$t = 0 \quad n$$

$$t = \infty \quad n(1 - \alpha) \quad m\alpha$$

$$\text{Initial vapour density} = D \propto \frac{1}{n}$$

Final vapour density at equilibrium

$$= d \propto \frac{1}{n(1 - \alpha) + m\alpha}$$

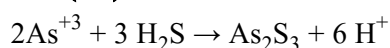
$$\frac{D}{d} = 1 - \alpha + m\alpha = 1 + \alpha(m - 1)$$

Since, we are given, slope = $\tan 45^\circ = 1$

$$\Rightarrow m - 1 = 1 \Rightarrow m = 2$$

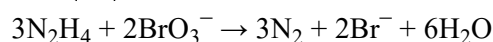
So, the equilibrium is as follows: $\text{A} \rightleftharpoons 2\text{B}$

11. **Ans (D)**



As_2S_3 is a negative sol and hence Ba^{+2} ions will precipitate the sol.

12. **Ans (A)**



13. **Ans (C)**

For chemical reactions. Heat absorbed at constant volume, is measured in a bomb calorimeter

$$q_v = C_v \times \Delta T = -20 \text{ kJ/K} \times (299 - 298)\text{K}$$

(-ve indicating the nature of reaction)

Thus ΔU for the combustion of 1 mol of graphite is

$$\frac{12 \text{ g mol}^{-1} \times (-20 \text{ kJ})}{2 \text{ g}} = -120 \text{ kJ/mol}$$

14. **Ans (B)**

All the compounds listed are nitrogen bases. A base donates its electron pair and the strength of the base is decided by the ease with which the species donates it which depends on the availability of electron density on the donating atom. The electron pair on nitrogen atom of compound (1) is not involved in any kind of resonance while nitrogen atoms of other molecules are involved in resonance. The electron pair on nitrogen atom of compound (4) is delocalized over two carbonyl groups and hence is least available. Among compounds (2) & (3), then electron density in the benzene ring is drained by the presence of $-\text{NO}_2$ group which in turn pulls electron density from the nitrogen atom. Hence, (2) is more basic than (3).

16. **Ans (D)**

For complex A

$$\text{EAN} = 36$$

$$\mu = 4.93 \text{ BM}$$

$$\text{O.S.} = +3$$

$$\text{hybridisation} \rightarrow sp^3 d^2.$$

The ligand is weak

For complex B

$$\text{EAN} = 37$$

$$\mu = 3.87 \text{ BM}$$

$$\therefore \text{ligand is weak and O.S. of Co} = +2$$

$$\text{Hybridisation} \rightarrow sp^3 d^2.$$

For complex C

$$\text{EAN} = 36$$

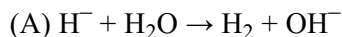
$$\mu = 0$$

$$\therefore \text{hybridisation} \rightarrow d^2 sp^3$$

ligand $\rightarrow$ strong.

$$\text{Oxidation state of Co} = +3$$

17. **Ans (D)**



(B) Due to decrease in stability of hydrides down the group in periodic table

18. **Ans (B)**

Data from NCERT

B $\rightarrow$ 801 kJ/mole

Tl $\rightarrow$ 589 kJ/mole

Ga $\rightarrow$ 579 kJ/mole

Al $\rightarrow$ 577 kJ/mole

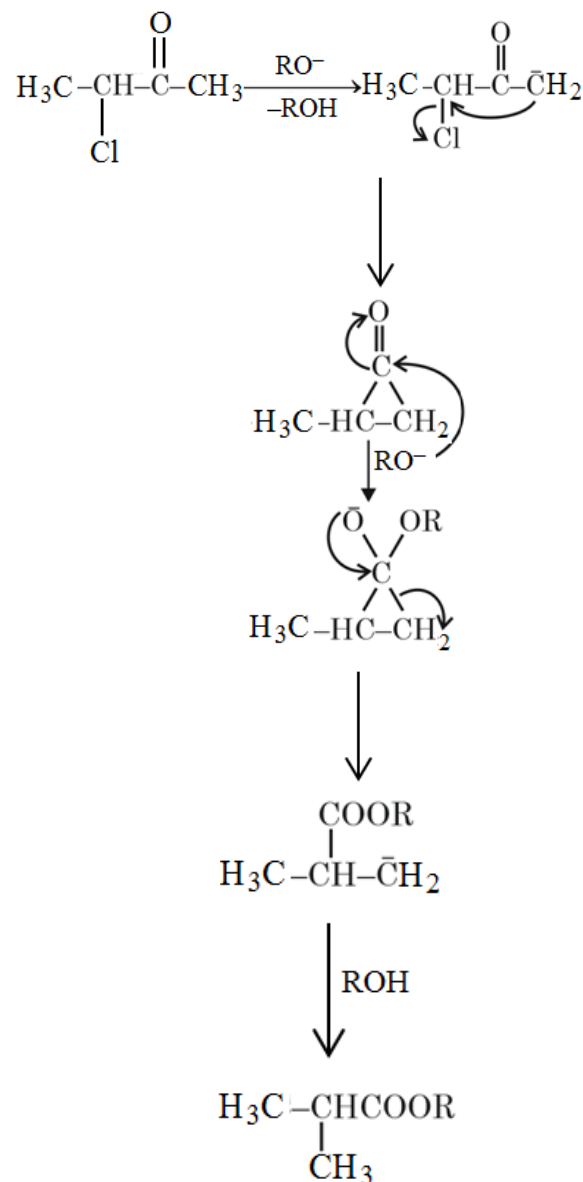
In $\rightarrow$ 558 kJ/mole

19. **Ans (D)**

$\text{SO}_2 \rightarrow$ bent

$\text{SeO}_2, \text{SiO}_2$, polymeric solid

20. **Ans (B)**



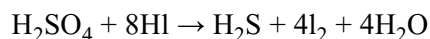
PART-2 : CHEMISTRY

SECTION-II

1. **Ans (9)**

$$\frac{\Delta E_{\text{Li}^{2+}} = 13.6 \times (3)^2 \text{ eV}}{\Delta E_{\text{H}} = 13.6 \times (1)^2 \text{ eV}} = 9$$

2. **Ans (4)**



3. **Ans (3)**

$$[\text{S}^{-2}] = \frac{10^{-17}}{0.1} = 10^{-16}$$

$$K = 10^{-21} = \frac{[\text{H}^+]^2 \times 10^{-16}}{0.1}$$

$$10^{-6} = [\text{H}^+]^2$$

$$[\text{H}^+] = 10^{-3}$$

$$\therefore \text{pH} = 3$$

4. **Ans (7)**

$$Z_{\text{B}^-} = 4 - 2 = 2$$

$$Z_{\text{A}^+} = 4$$

$$Z_{\text{Z}^{2-}} = 1$$

$\therefore \text{A}_4\text{B}_2\text{Z}$ is the formula unit.

$$x + y + c = 7$$

5. **Ans (3)**

$$\pi r^2 = 549 \text{ \AA}^2$$

$$2\pi r = 83.05 \text{ \AA}$$

$$r = 13.22 \text{ \AA}$$

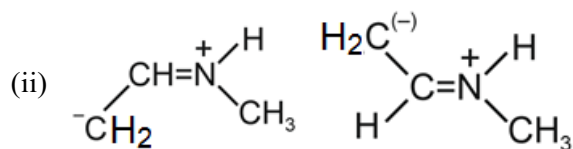
$$\frac{0.529 \times n^2}{Z} = 13.22$$

$$n = 5$$

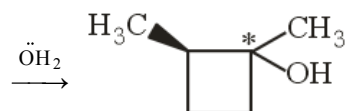
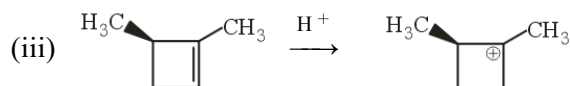
Number of maximum lines possible is = 3

6. Ans (8)

P = 4



∴ Q = 2



2 stereoisomers

∴ R = 2

P + Q + R = 4 + 2 + 2 = 8

7. Ans (3)

M[ABCD] type complex show three geometrical isomers.

8. Ans (4)

$$X_{\text{CS}_2} = \frac{0.001}{0.001 + 1} = 0.001$$

$$X_{\text{acetone}} = 1$$

$$P_{\text{acetone}} = 1 \times 399 = 399 \text{ mm of Hg}$$

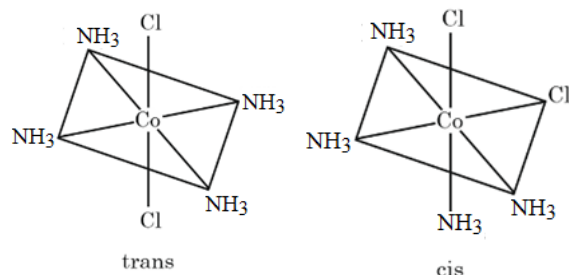
$$P_{\text{CS}_2} = 0.001 \times 10^3 = 1 \text{ mm of Hg}$$

$$P_{\text{Total}} = P_{\text{acetone}} + P_{\text{CS}_2} = (399 + 1) = 400 \text{ mm of Hg}$$

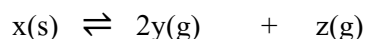
9. Ans (2)

$$0.93 = i \times 1.86 \times 0.25 \Rightarrow i = 2$$

Hence, the formula of complex is $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$



10. Ans (9)



case I 2P P

case II $K_p = 4p^3$
 $2p - 0.2p \quad p + 1 - 0.1p$

$$K_p = (1.8p)^2 (0.9p + 1)$$

$$\Rightarrow 4p^3 \Rightarrow p = 2.988 \text{ bar}$$

$$p \text{ in case II} = 2.7p + 1 = \approx 9$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

Range of $a^2 \sin^2 \theta + b^2 \operatorname{cosec}^2 \theta$ is

$$\begin{cases} [2ab, \infty) & \text{if } a \geq b \\ [a^2 + b^2, \infty) & \text{if } a < b \end{cases}$$

Here $b > a$, so range is $[41, \infty)$

2. Ans (D)

Taking $n!$ common from R_1 , $(n+1)!$ from

R_2 and $(n+2)!$ from R_3 , we obtain

$$\Delta_n = n! (n+1)! (n+2)! \Delta$$

$$\text{where } \Delta = \begin{vmatrix} 1 & n+1 & (n+1)(n+2) \\ 1 & n+2 & (n+2)(n+3) \\ 1 & n+3 & (n+3)(n+4) \end{vmatrix}$$

using $R_3 \rightarrow R_3 - R_2$ and $R_2 \rightarrow R_2 - R_1$, we get

$$\Delta = \begin{vmatrix} 1 & n+1 & (n+1)(n+2) \\ 0 & 1 & 2(n+2) \\ 0 & 1 & 2(n+3) \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2n+4 \\ 1 & 2n+6 \end{vmatrix} = 2$$

Thus, $\Delta_n = 2n! (n+1)! (n+2)!$

$$\Rightarrow \Delta_{n+1} = 2(n+1)! (n+2)! (n+3)!$$

$$\text{Now, } \frac{(3n^3 - 5) \Delta_n}{\Delta_{n+1}} = \frac{(3n^3 - 5)}{(n+1)(n+2)(n+3)}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{(3n^3 - 5) \Delta_n}{\Delta_{n+1}}$$

$$= \lim_{n \rightarrow \infty} \frac{\left(3 - \frac{5}{n^3}\right)}{\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \left(1 + \frac{3}{n}\right)} = 3$$

3. **Ans (C)**

$$a - 2b + 10 > 0$$

$$\Rightarrow 1 \leq b < (a + 10)/2$$

$$\text{Thus, } b \in [1, (a + 10)/2]$$

$$\text{When } a = 1, 2, \quad b \in [1, 5]$$

$$\text{When } a = 3, 4, \quad b \in [1, 6]$$

$$\text{When } a = 5, 6, \quad b \in [1, 7]$$

$$\text{When } a = 7, 8, \quad b \in [1, 8]$$

$$\text{When } a = 9 \quad b \in [1, 9]$$

Thus, then inequality $a - 2b + 10 > 0$ is satisfied for $2 \times 5 + 2 \times 6 + 2 \times 7 + 2 \times 8 + 9 = 61$ ordered pairs (a, b).

Total number of ways of choosing (a, b) is

$$9 \times 9 = 81$$

$$\therefore \text{required probability is } \frac{61}{81}$$

4. **Ans (C)**

$$A + B = AB \Rightarrow A = AB - B = (A - I)B.$$

Pre-multiplying by A^{-1} we get

$$I = A^{-1}A = (I - A^{-1})B$$

$$\Rightarrow I - A^{-1} \text{ is invertible.}$$

5. **Ans (B)**

For $|x| < 1$, $x^{2n} \rightarrow 0$ as $n \rightarrow \infty$ and for $|x| > 1$,

$1/x^{2n} \rightarrow 0$ as $n \rightarrow \infty$. So

$$f(x) = \begin{cases} \ln(2+x) & \text{if } |x| < 1 \\ \lim_{n \rightarrow \infty} \frac{x^{-2n} \ln(2+x) - \sin x}{x^{-2n} + 1} = -\sin x & \text{if } |x| > 1 \\ \frac{1}{2}[\ln(2+x) - \sin x] & \text{if } |x| = 1 \end{cases}$$

$$\text{Thus } \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (-\sin x) = -\sin 1$$

$$\text{and } \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \log(2+x) = \log 3.$$

6. **Ans (D)**

Required image lies on the line through $A(-1, 3, 4)$

and perpendicular to $x - 2y = 0$ that is on the line

$$\frac{x+1}{1} = \frac{y-3}{-2} = \frac{z-4}{0} = t(\text{say})$$

Let the coordinates of image point be

$$B(t-1, -2t+3, 4)$$

$$\text{Mid point M of AB is } \left(\frac{t-2}{2}, \frac{-2t+6}{2}, 4 \right)$$

which lies on the plane $x - 2y = 0$

$$\Rightarrow \frac{1}{2}t - 1 + 2t - 6 = 0 \Rightarrow t = \frac{14}{5}$$

$$\text{Required image is } \left(\frac{9}{5}, -\frac{13}{5}, 4 \right)$$

7. **Ans (D)**

Given equation can be written as $y' + \frac{1}{x}y = \frac{1}{x}e^x$,

which is a linear equation whose I.F. is

$$e^{\int \frac{1}{x} dx} = e^{\log x} = x. \text{ Multiplying with I.F. we}$$

have

$$\int \frac{d}{dx}(yx) = \int e^x$$

$$yx = e^x + C$$

$$\text{Put } x = a, y = b, \text{ we have } ab - e^a = C$$

$$\text{Thus } y = \frac{e^x + ab - e^a}{x}$$

$$\lim_{x \rightarrow 1} y(x) = \lim_{x \rightarrow 1} \frac{e^x + ab - e^a}{x} = e + ab - e^a.$$

8. **Ans (C)**

Note that $|a| = |b| = |c| = |d| = 1$.

$$\text{Also, } |2a - 9d|^2 = |-6b - 7c|^2$$

$$\Rightarrow 4|a|^2 + 81|d|^2 - 36a \cdot d = 36|b|^2 + 49|c|^2 + 84b \cdot c$$

$$\Rightarrow -3(\cos \alpha \cos \delta - \sin \alpha \sin \delta) = 7(\cos \beta \cos \gamma - \sin \beta \sin \gamma)$$

$$\Rightarrow 3\cos(\alpha + \delta) + 7\cos(\beta + \gamma) = 0$$

9. **Ans (A)**

$$p \rightarrow (q \rightarrow r) \equiv \sim p \vee (q \rightarrow r)$$

$$\equiv \sim p \vee (\sim q \vee r)$$

$$\equiv [(\sim p) \vee (\sim q)] \vee r$$

$$\equiv \sim(p \wedge q) \vee r$$

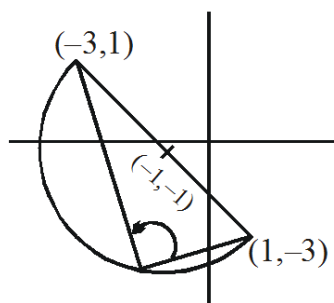
$$\equiv p \wedge q \rightarrow r$$

$\therefore$ given statement is a tautology.

10. Ans (D)

The given conditions can be expressed as
 $n(M \cup P \cup C) = 50$, $n(M) = 37$, $n(P) = 24$,
 $n(C) = 43$, $n(M \cap P) \leq 19$,
 $n(M \cap C) \leq 29$ and $n(P \cap C) \leq 20$,
 $n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(M \cap C) - n(P \cap C) + n(M \cap P \cap C)$
 $\Rightarrow 50 = 37 + 24 + 43 - n(M \cap P) - n(M \cap C) - n(P \cap C) + n(M \cap P \cap C)$
 $\Rightarrow n(M \cap P \cap C) \leq n(M \cap P) + n(M \cap C) + n(P \cap C) - 54$.
 Therefore, the number of students that could have passed all three exams is at most
 $19 + 29 + 20 - 54 = 14$.

11. Ans (D)



$$\arg\left(\frac{z - (-3 + i)}{z - (1 - 3i)}\right) = \frac{\pi}{2}$$

So the required locus is shown in diagram with centre $(-1, -1)$ and radius $2\sqrt{2}$.

12. Ans (C)

Case-I : $\frac{x+3}{3} = \frac{2}{x+1}$

$$\Rightarrow x^2 + 4x - 3 = 0$$

$$\Rightarrow x = \frac{-4 + \sqrt{28}}{2} \text{ or } \frac{-4 - \sqrt{28}}{2} \text{ (reject).}$$

Case-II : $\frac{x+3}{3} = \frac{x+1}{2} \Rightarrow x = 3$

13. Ans (B)

$$\therefore d \neq 0$$

Lines must be parallel

$$\Rightarrow \frac{k}{k-1} = \frac{k-1}{k} \neq 1 \Rightarrow k^2 = k^2 - 2k + 1$$

$$\Rightarrow k = \frac{1}{2}$$

$$\Rightarrow \text{Lines are } x - y = 2 \text{ \& } x - y = -2$$

$$\Rightarrow d = \frac{4}{\sqrt{2}} \Rightarrow d^2 = 8$$

14. Ans (A)

If we mark any odd number or even number stones, there are always odd number of stones in between them

$$\text{So, } {}^8C_5 + {}^7C_5 = 77$$

15. Ans (C)

$$\sqrt{1 - \sin x} = \cos x$$

Squaring

$$1 - s = 1 - s^2 \text{ (where } s = \sin x)$$

$$\Rightarrow s = 0, 1$$

$$\Rightarrow x = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi, \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}$$

but $x = \pi, 3\pi, 5\pi$ are rejected

$$\Rightarrow \text{Number of solutions} = 6$$

16. Ans (B)

$$\text{Solving } y^2 = x \text{ \& } 5x + 2y = 1 + 2x^2$$

$$(y+1)^2(2y^2 - 4y + 1) = 0$$

$y = -1$ is a repeated root

$$\Rightarrow \text{point of contact is } (1, -1)$$

$$\therefore \text{common tangent is } y(-1) = \frac{x+1}{2}$$

$$\Rightarrow x + 2y + 1 = 0$$

17. Ans (A)

Limit

$$\lim_{x \rightarrow 0} \left[\frac{c \sin cx}{2} + \frac{x(\sqrt{x+a^2}+a)}{4(\sqrt{x+a^2}-a)(\sqrt{x+a^2}+a)} \right]$$

$$\frac{c}{2} + \frac{2a}{4} = \frac{a+c}{2}$$

$$f(0) = b$$

$$\text{function is continuous then } f(0) = b = \frac{(a+c)}{2}$$

18. Ans (A)

$$f(x) = f\left(\frac{x}{2}\right) = f\left(\frac{x}{2^2}\right) = \dots$$

$$\Rightarrow f(x) = f\left(\frac{x}{2^n}\right) \Rightarrow \lim_{n \rightarrow \infty} f(x) = \lim_{n \rightarrow \infty} f\left(\frac{x}{2^n}\right)$$

$$f(x) = f(0) = \text{constant}$$

$\therefore f$ is continuous

$$\Rightarrow f(x) = f(2012) = \frac{\pi}{2}$$

$$\Rightarrow L = \lim_{x \rightarrow 0} \frac{\cos^2 \pi/2 + 1 - \sin^3 \pi/2}{\sin^2 x} = 0$$

19. Ans (C)

$$y^2 = x(2-x)^2$$

$$\Rightarrow 2yy' = (2-x)^2 + 2x(x-2)$$

$$\Rightarrow y' = \frac{3x^2 - 8x + 4}{2y} \Rightarrow y'_{(1,1)} = -\frac{1}{2}$$

$$\Rightarrow \text{Equation of tangent is } y - 1 = -\frac{1}{2}(x - 1)$$

$$\Rightarrow 2y - 2 = 1 - x \Rightarrow x + 2y = 3$$

$$\text{Solve with } y^2 = x(x-2)^2$$

$$\Rightarrow (x-3)^2 = 4x(x-2)^2$$

$$\Rightarrow x^2 - 6x + 9 = 4x^3 - 16x^2 + 16x$$

$$\Rightarrow 4x^3 - 17x^2 + 22x - 9 = 0$$

$$\Rightarrow (x-1)^2(4x-9) = 0$$

$$\Rightarrow x = \frac{9}{4} \Rightarrow y = \pm \frac{3}{8}$$

$$\Rightarrow \left| \frac{a}{b} \right| = \frac{9 \times 8}{4 \times 3} = 6$$

20. Ans (B)

$$\sigma^2 = \frac{n_1 \sigma_1^2 + n_2 \sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2} (\bar{x}_1 - \bar{x}_2)^2$$

$$\sigma^2 = \frac{6 \times 24 + 3 \times 36}{6 + 3} + \frac{6 \times 3}{(6 + 3)^2} (11 - 14)^2$$

$$\sigma^2 = \frac{144 + 108}{9} + \frac{18}{81} \times 9$$

$$= 28 + 2 = 30$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (7)

$$x^2 + 20 \leq 9x \Rightarrow x^2 - 9x + 20 \leq 0$$

$$\Rightarrow (x-4)(x-5) \leq 0$$

$$\Rightarrow A = [4, 5]$$

$$f'(x) = 6x^2 - 30x + 36$$

$$= 6(x-2)(x-3) > 0 \text{ on } [4, 5]$$

Thus f increases on $[4, 5]$, Therefore,

$$\max_{x \in A} f(x) = f(5) = 7.$$

2. Ans (1)

$$F(x) = \int f(x) dx = x + 3^x + C \text{ but } F(2) = 7 \text{ so}$$

$$7 = 2 + 9 + C$$

$$\Rightarrow C = -4. \text{ Hence } F(x) = x + 3^x - 4. \text{ Now } F(x) = 0$$

$$\Rightarrow x = 1.$$

3. Ans (5)

$$f(x) = \left(2 + \frac{\cos^{-1} x + \sin^{-1} x}{\pi}\right) \left(\frac{\cos^{-1} x - \sin^{-1} x}{\pi}\right)$$

$$f(x) = \left(2 + \frac{\cos^{-1} x + \sin^{-1} x}{\pi}\right) \left(\frac{2\cos^{-1} x}{\pi} - \frac{1}{2}\right)$$

$$\text{greatest value} = \left(2 + \frac{1}{2}\right) \left(2 - \frac{1}{2}\right) = \frac{15}{4}$$

$$\text{least value} = \frac{5}{2} \times \left(-\frac{1}{2}\right) = -\frac{5}{4}$$

$$\text{difference} = \frac{15}{4} + \frac{5}{4} = 5$$

4. Ans (3)

$$T_{r+1} = {}^{40}C_r x^{\frac{40-r}{3}} \cdot x^{\frac{2r}{5}}$$

$$= {}^{40}C_r x^{\frac{200+r}{15}}$$

$$r = 10, 25, 40$$

$$\Rightarrow \text{Number of such terms} = 3$$

5. Ans (49)

$$S = (2-d) \left(\frac{2}{3} \right) + (2+d) \left(\frac{4}{9} \right) + (2+3d) \left(\frac{8}{27} \right) + \dots$$

$$S \left(\frac{2}{3} \right) = (2-d) \frac{4}{9} + (2+d) \frac{8}{27} + \dots$$

Subtracting

$$\frac{S}{3} = (2-d) \frac{2}{3} + 2d \left(\frac{4}{9} \right) + 2d \left(\frac{8}{27} \right) + \dots$$

$$= (2-d) \frac{2}{3} + 2d \left[\frac{4}{9} + \frac{8}{27} + \dots \right]$$

$$= (2-d) \frac{2}{3} + 2d \left[\frac{4}{9} \cdot \left(\frac{3}{1} \right) \right] = \frac{4}{3} + 2d$$

$$\Rightarrow S = 4 + 6d$$

$$\text{Given } S = \frac{5}{2} - 2 = \frac{1}{2} = 4 + 6d$$

$$\Rightarrow d = -\frac{7}{12}$$

6. Ans (6)

$$AC = 5 = 2ae$$

$$\Rightarrow 2ae = 5 \dots (1)$$

$$\text{Also } AB + BC = 2a$$

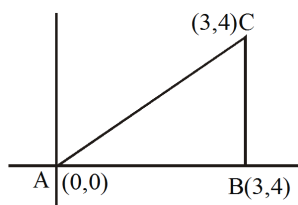
$$\Rightarrow 2a = 7$$

$$\Rightarrow a = \frac{7}{2}$$

$$\therefore e = \frac{5}{7} \Rightarrow 1 - \frac{b^2}{a^2} = \frac{25}{49} \Rightarrow b = \sqrt{6}$$

$$\text{Area} = \pi ab = \pi \times \frac{7}{2} \times \sqrt{6}$$

$$\therefore P = 6$$



7. Ans (3)

$$\text{Let the point be } (\alpha, \beta) \Rightarrow \beta = \alpha + 2$$

Chord of contact of hyperbola is $T = 0$

$$\Rightarrow \frac{\alpha x}{6} - \frac{\beta y}{2} = 1 \Rightarrow \frac{\alpha x}{6} - \frac{(\alpha + 2)y}{2} = 1$$

$$\Rightarrow \alpha \left(\frac{x}{6} - \frac{y}{2} \right) - \left(\frac{2y}{2} + 1 \right) = 0$$

Since this passes through (λ, μ)

$$\therefore \frac{\lambda}{6} - \frac{\mu}{2} = 0 \text{ and } \frac{2\mu}{2} + 1 = 0 \Rightarrow \frac{\lambda}{\mu} = 3$$

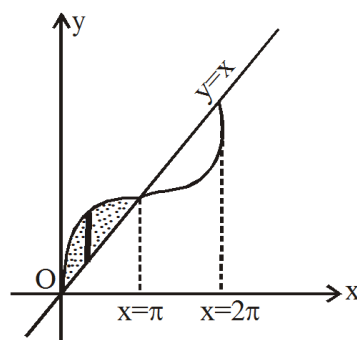
8. Ans (150)

$$x^2 + y^2 = \left(t + \frac{1}{t} \right)^2 - 2$$

$$x^2 + y^2 = (x+y)^2 - 2$$

$$xy = 1 \Rightarrow y = \frac{1}{x} \Rightarrow \frac{dy}{dx} = -\frac{1}{x^2}$$

9. Ans (2)



For the points of intersection of the given curves

$$x = x + \sin x$$

$$\Rightarrow \sin x = 0$$

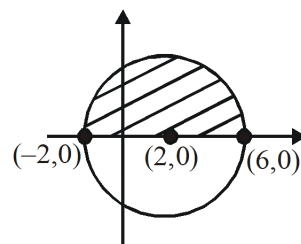
$$\Rightarrow x = 0, \pi$$

$\therefore$ required area

$$= \int_0^\pi [(x + \sin x) - x] dx$$

$$= \int_0^\pi \sin x dx = -[\cos x]_0^\pi = 2$$

10. Ans (8)



$$\int_{-2}^6 \sqrt{(4)^2 - (x-2)^2} dx$$

$$(x-2)^2 + y^2 = (4)^2$$

Area of semicircle

$$\frac{\pi(4)^2}{2} = 8\pi$$